National Institute of Business Management

Higher National Diploma in Information Systems

HNDIS25.1F

Agile and DevOps

Coursework Report

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1. Project Introduction & Requirements Analysis

1.1 Introduction

With the rapid growth of online recruitment platforms, job seekers increasingly depend on digital platforms such as LinkedIn to search for employment opportunities. While these platforms have made job searching easier and more accessible, they have also opened doors for fraudulent activities. Fake job postings are becoming increasingly common, with scammers often targeting job seekers by advertising misleading or completely fake vacancies. These scams can cause serious problems, including financial loss, identity theft, and emotional stress.

To tackle this issue, our project focuses on creating an innovative software solution that can identify and highlight fake job postings, helping users stay while searching for jobs. The system aims to improve user safety by defecting suspicious job listings and providing warnings before users engage with them.

1.2 Background of Fake Job Scams

Fake jobs scams have become increasingly common with the rise of online professional networking platforms. Scammers often create fake company profiles or impersonate real organizations to attract job seekers. These fake postings usually promise high salaries, remote work opportunities, or quick hiring processes, which appear attractive to users.

In many cases, victims are asked to share personal information, make payments for training, or communicate outside the platform. Due to the lack of awareness and proper verification mechanisms, many job seekers fall into these traps. This highlights the urgent need for a system that can automatically detect suspicious job postings and alert users in advance.

1.3 Problem Statement

Although platforms like LinkedIn have policies to prevent fraud, fake job postings still appear frequently. Job seekers often find it difficult to identify whether a job post if legitimate or fraudulent. The absence of an intelligent detection system puts users at risk of scams, financial loss, and misuse of personal information.

Therefore, there is a strong need for a software solution that can analyze job postings, identify potential fraud indicators, and provide early warnings to users to ensure a safe job-search experience.

1.4 Objectives of the System

The main objectives of the proposed system are:

- To identify and defect fake job postings using intelligent analysis
- To protect job seekers from online job-related scams
- To analyze job descriptions and recruiter information for suspicious patterns
- To provide alerts or risk indicators for suspicious job listings
- To increase trust and safety within online recruitment platforms

1.5 Scope of the Project

The scope of this project includes the design and development of a software system that can:

- Analyze job descriptions and recruiter profiles
- Detect suspicious keywords, unrealistic offers, and abnormal patterns
- Provide a risk score or warning level for each job post
- Allow users to report fake or suspicious job advertisements
- Work as a supportive tool for job seekers rather than replacing existing platforms

The project will focus on detection and warning mechanisms and will not involve direct interaction with payment systems or personal financial data.

Requirements Management

Effective requirements management is essential to ensure the success of the system. The project follows Agile practices to identify, prioritize, and manage requirements throughout development.

Identified EPICs

EPIC ID	EPIC Name	Description
EPIC 1	Job Data Analysis	Analyze job postings to identify suspicious content and patterns
EPIC 2	Fraud Detection	Detect fake job listings using predefined rules and logic

EPIC 3	User Interaction	Display warnings, alerts, and allow user feedback
EPIC 4	System Management	Manage data, reports, and system performance

Minimally Releasable Features (MRFs)

EPIC	MRF ID	MRF Description
Job Data Analysis	MRF 1.1	Extract job details such as title, description, and company name
Job Data Analysis	MRF 1.2	Identify keywords and phrases commonly used in fake job posts
Fraud Detection	MRF 2.1	Detect unrealistic salary ranges and suspicious offers
Fraud Detection	MRF 2.2	Assign a risk score to each job post
User Interaction	MRF 3.1	Display warning messages for high-risk jobs
User Interaction	MRF 3.2	Enable users to easily report job listings that appear suspicious or untrustworthy
System Management	MRF 4.1	Generate basic reports for system monitoring

2. User Stories and Agile Planning

2.1 User Stories and Estimation

Highest Priority Minimally Releasable Features (MRFs) are broken down into User Stories as part of requirements management.

The highest priority features were chosen and divided into user stories based on the identified EPICs and Minimally Releasable Features (MRFs). User stories guarantee that development is focused on providing value to end users by describing system requirements from the viewpoint of the user.

User Stories

- **User Story 01:**

As a job seeker, I want to submit a job posting to the system, so that I can determine whether it is authentic or a scam.

- **User Story 02:**

As a job seeker, I want the system to analyze job descriptions using pre-established scam indicators, so that I can identify questionable job postings early on.

- **User Story 03:**

As a job seeker, I want to see a trust score for every job posting, so that I can quickly assess the degree of risk before submitting an application.

- **User Story 04:**

As a job seeker, I want to receive warning messages for high-risk job advertisements, so that I can prevent falling for scams.

- **User Story 05:**

As a user, I want to report suspicious job advertisements, so that I can help the system increase the accuracy of its detection.

- **User Story 06:**

As an administrator, I want to oversee and update scam detection rules, so that I can keep the system effective against emerging scam patterns.

2.2 Planning Poker for User Story Sizing

Planning Poker was used to estimate the amount of work required for each user story. This technique allows team members to individually assess a story's complexity using the Fibonacci sequence values (1, 3, 5, 8, etc.). This method enables team members to independently estimate the complexity of a story. A collective review was conducted, after which a consensus was reached to ensure accurate and reasonable estimates.

User Story ID	Description	Story Points
US-01	Send in your job posting for review.	5

User Story ID	Description	Story Points
US-02	Use scam indicators for analyzing the job description.	3
US-03	Create a trust score for job advertisements	8
US-04	Show warns indicators for risky positions	5
US-05	Report any questionable job postings.	3
US-06	Administration of rules for scam detection	8

An explanation of estimation

Simple features like warning display received lower story points, while more complex features like rule management and trust score estimation received higher story points. The team was able to effectively balance workload within each sprint and identify possible risks thanks to Planning Poker.

The team made sure that the requirements were clear and that the sprint planning was efficient by breaking down MRFs into clear user stories and estimating them with Planning Poker. This approach facilitates development and increases the overall success of the Fake Job & Internship Detection Platform.

3. System Design & Diagrams

3.1 Introduction to System Design

- The system design phase defines the understanding of the Fake Job & Internship Detection Platform. This section presents the Design, workflows, and database structure of the proposed system using standardized diagrams.

3.2 System Architecture Diagram

3.2.1 Diagram Description

- The System Architecture Diagram shows technical structure of the Fake Job & Internship Detection Platform. The architecture is designed in layers of presentation, application, and data, allowing clear separation of responsibilities and supporting system scalability.

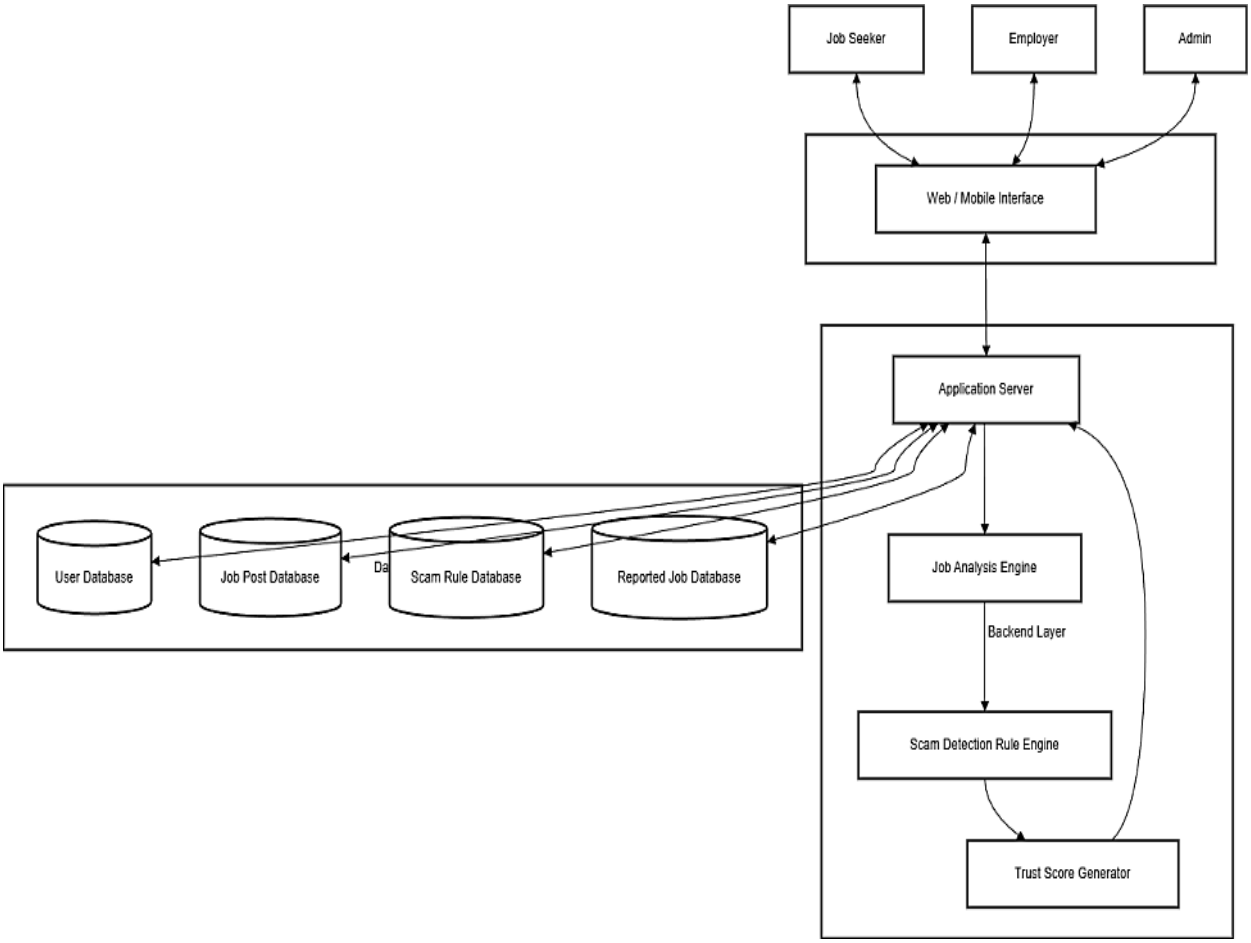


Figure 01 - [System Architecture Diagram]

3.2.2 Architecture Explanation (System Workflow)

- Job seekers, employers, and administrators access the system via a web or mobile interface.
- Job data submitted by employers is processed by the application server.
- The job analysis engine examines job descriptions.
- Scam detection rules are applied using the rule engine.
- A trust score is generated based on predefined indicators.
- Results and warnings are displayed to job seekers.
- All relevant data is stored in centralized databases for consistency.

3.3 Use Case Diagram

3.3.1 Diagram Description

- The Use Case Diagram represents the Identify the actors along with their interactions and the Fake Job & Internship Detection Platform system.

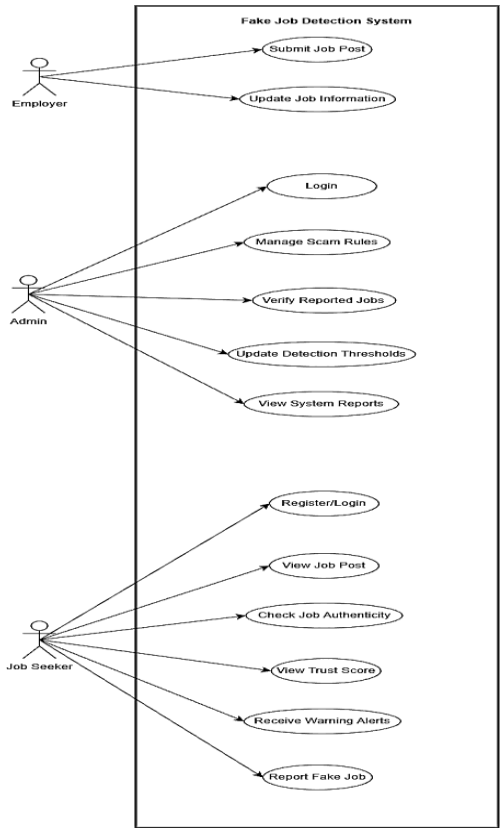


Figure 02 - [Use Case Diagram]

3.3.2 Use Case Explanation

- Job seekers can view job posts, check job authenticity, receive trust scores, and report suspicious postings.
- Administrators manage scam detection rules, verify reported jobs, and analyze system reports.
- Employers submit and manage job postings.

3.4 Data Flow Diagram

3.4.1 Data Flow Diagram (Level 0)

Diagram Description

- The Level 0 Data Flow Diagram presents the entire system as a single process and shows interactions between external entities and the system.

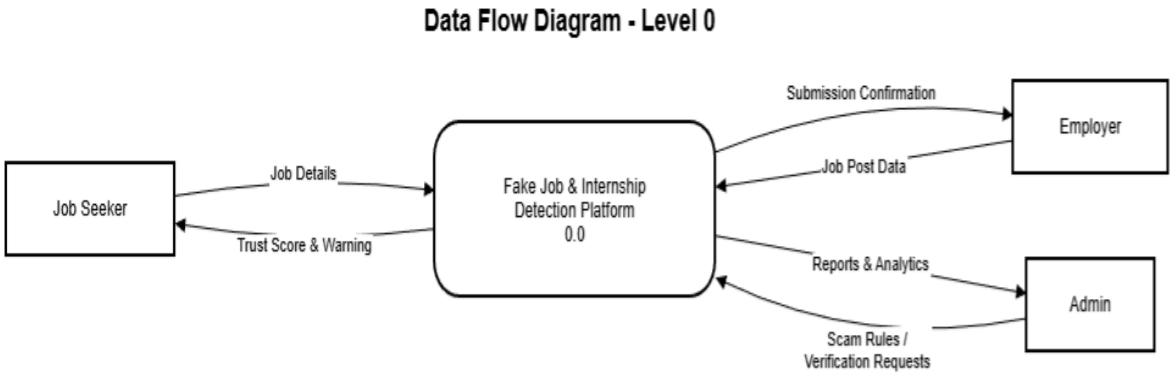


Figure 03 - [Data Flow Diagram]

Explanation

- Job seekers submit job details for authenticity verification and receive trust scores.
- Employers submit job postings to the system.

- Administrators provide scam detection rules and receive analytical reports.

3.4.2 Data Flow Diagram (Level 1)

Diagram Description

- The Level 1 Data Flow Diagram decomposes the main system into multiple subprocesses.

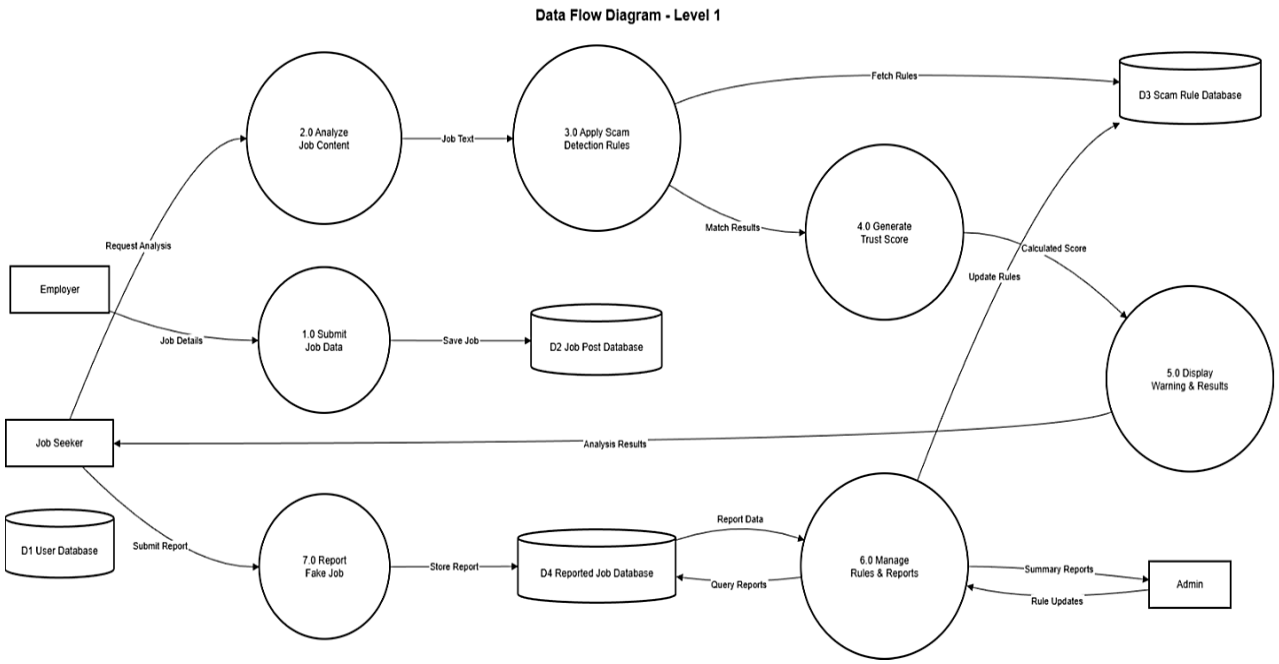


Figure 04 - [Data Flow Diagram]

Explanation

- Job data is collected and stored.
- Scam detection rules are applied.

- scores are calculated.
- Results are displayed to users.
- Reports and feedback are stored for administrative review.

3.5 Entity Relationship Diagram (ERD)

3.5.1 Diagram Description

- The Entity Relationship Diagram (ERD) defines the database structure of the Fake Job & Internship Detection Platform. It identifies key entities, their attributes, and the relationships between them.

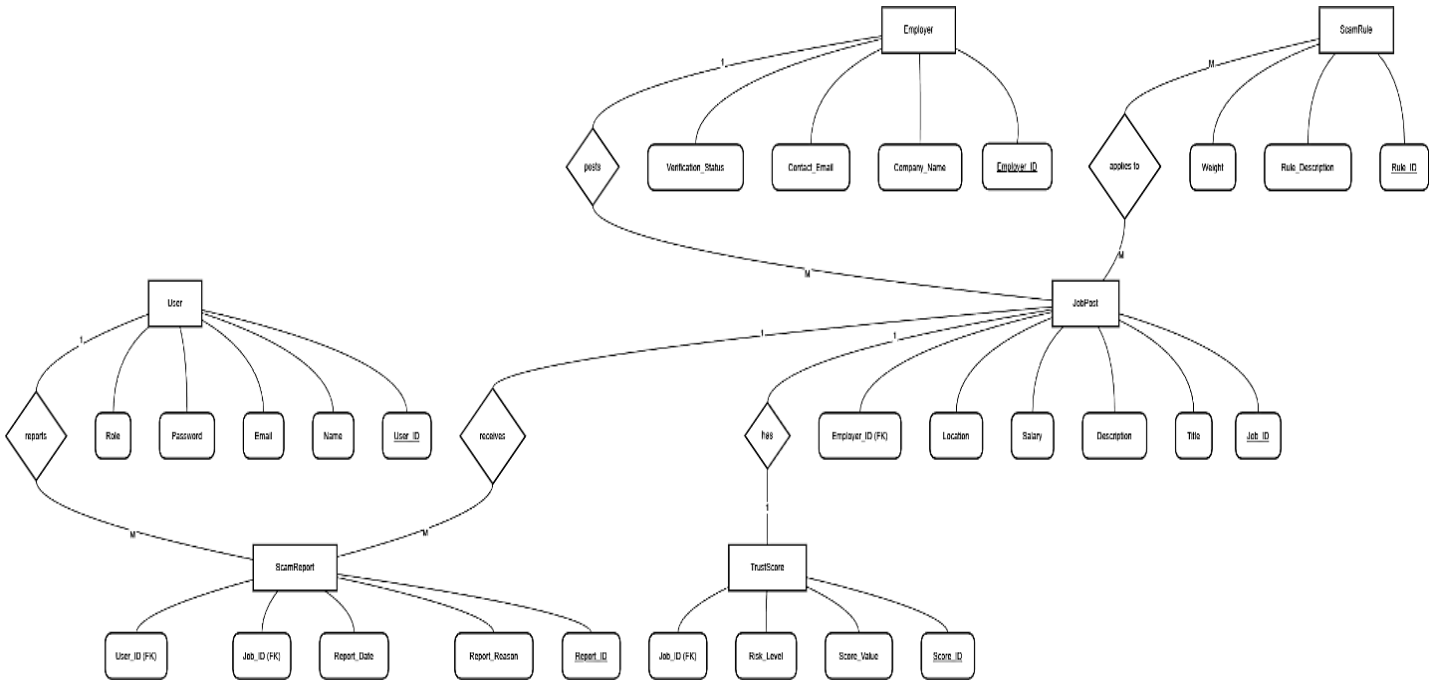


Figure 05 - [ER Diagram]

Explanation

- Users interact with job postings and submit reports.
- Employers create multiple job postings.

- Each job post generates a trust score.
- Scam reports link users and job postings.
- Scam rules support decision-making during job analysis.

3.6 Conclusion of System Design

- The system design effectively supports the objectives of the Fake Job & Internship Detection Platform by providing clear functional interactions, efficient data flow, and a well-organized database model.

4. Agile Execution

Agile execution focuses on continuously delivering value throughout the Software Development Life Cycle (SDLC). It emphasizes flexibility and the ability to adapt requirements to meet business objectives, deliver effective results, and ensure customer success. Effective communication among all team members is essential, as it ensures everyone clearly understands their roles, responsibilities, and overall workflow.

This approach is implemented through Agile practices such as Daily Scrum meetings, Sprint Reviews, and Sprint Retrospectives, ensuring that customer needs are met while adding value to the business and strengthening the organization's reputation. In our platform, every team member shares responsibility and works collaboratively to develop the product and deliver value to users.

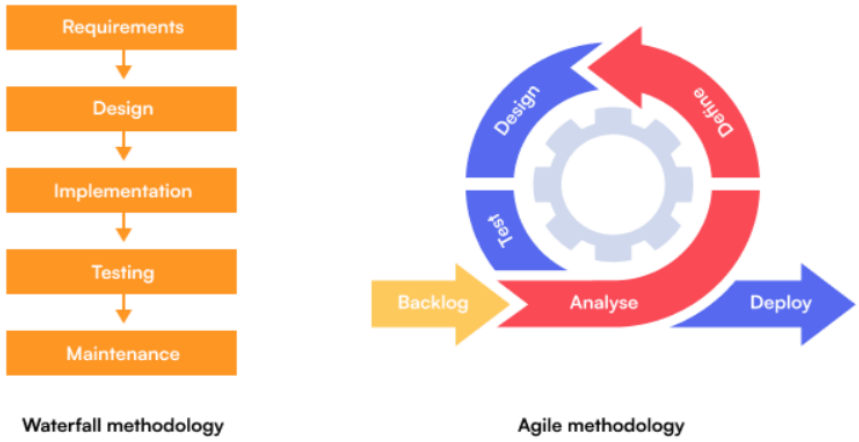
4.1 Agile Methodology

Agile methodology is widely used across various industries and business domains due to its ability to enhance customer satisfaction in today's fast-changing environment. It has transformed modern project management by promoting adaptability, collaboration, and continuous improvement.

Agile replaces rigid and lengthy planning with an iterative development process that allows teams to respond quickly to change. It encourages motivated teams to deliver solutions faster, continuously improve product quality, and maintain close collaboration with customers to ensure project success.

Our platform follows Agile practices to provide better user experience. Initially, the fake job and internship detection platform was developed using the Waterfall model. However, after

transitioning to Agile methodology, product efficiency improved significantly and the number of defects was reduced. Agile also helps meet customer needs effectively by optimizing resource usage and minimizing risks.



Testsigma (August 6, 2025) Agile vs waterfall: Difference between methodologies

4.2 Daily Scrum

The purpose of the Daily Scrum is to track progress toward Sprint goals and manage the Sprint Backlog effectively. It is a short meeting, usually lasting 10–15 minutes, conducted every morning by the Scrum team. These meetings can be held in person or online and are typically scheduled at the same time each working day during the Sprint.

During the meeting, developers share updates on their progress, explain the methods they are using, and plan tasks for the next day. This process supports self-management, improves teamwork, and strengthens communication within the team.

Daily Scrums also help with better decision-making, idea sharing, and workload reduction. If needed, team members may meet again later in the day to discuss adjustments or plan remaining Sprint work.

Common discussion points include:

- What have you worked on since the last meeting?
- What do you plan to finish today?
- Are there any problems or implements to your progress?

For instance, the data analysis team might review how the platform's requirements were collected, such as examining job registration patterns in Sri Lanka or trends in internship hiring.

4.3 Sprint Review

The Sprint Review is conducted to evaluate the outcomes of the Sprint and decide on future improvements. It is a collaborative session where stakeholders inspect completed work and identify opportunities for enhancement.

The Product Owner ensures that relevant stakeholders—such as the development team, customers, and job providers—are invited. During the session, product features are demonstrated, allowing participants to experience the system firsthand and provide valuable feedback.

All feedback is collected, grouped, and discussed by the team to identify the most valuable features or improvements to prioritize in future Sprints.

4.4 Sprint Retrospective

(Source: Scrum.org, YouTube, n.d.)

The Sprint Retrospective focuses on improving team effectiveness and product quality. It allows the team to reflect on how the Sprint went in terms of collaboration, processes, tools, and overall performance.

Having a collaborative environment encourages open discussions where team members can identify better approaches for testing, research, and development. This process helps uncover bugs and improve platform quality.

The team may use techniques such as the "Perfection Game" to suggest and evaluate improvement ideas. Members vote on the most impactful improvement, which is then added to the backlog for

the next Sprint. The session concludes by appreciating everyone's contributions, reinforcing teamwork and continuous improvement.

Reference

Scrum.org - (Scrum.org, n.d.)

(Omonije, 2022)

5. DevOps Tools and Technologies

For the Fake Job & Internship Detection Platform, our team moved beyond traditional development methods and adopted a DevOps approach. Since online scams evolve rapidly, we realized we couldn't afford to wait weeks to release updates. DevOps allowed us to combine our development and operations efforts into a single, smooth workflow.

By integrating tools like Jenkins and Docker, we automated repetitive tasks. This meant that instead of spending hours manually installing software or fixing server issues, we could focus on improving the scam detection algorithms. This approach ensured that our platform remained stable, reliable, and capable of handling user traffic from day one.

5.1 Benefits of Using DevOps in This Project

1. Faster Development & Deployment: Scammers frequently change the way they phrase job postings to deceive users. DevOps practices, we can quickly update detection rules and deploy them to the live system, ensuring users remain protected in real time.
2. Improved Software Quality: By using automated testing, we catch bugs instantly. If a developer accidentally breaks the "Search" feature, the system warns us before the code goes live.
3. Better Collaboration: In a traditional setup, developers and testers work in silos. In this project, tools like Git and Jira kept everyone on the same page, preventing version conflicts.
4. Scalability & Reliability: As the number of job seekers using our platform grows, our cloud setup ensures the site doesn't crash under the load.
5. Reduced Downtime: Automated monitoring means we know if the site is down before the users do, allowing us to fix issues immediately.

5.2 DevOps Tools Selected

DevOps Area	Tool	Purpose in Project
Version Control	Git / GitHub	To manage our source code and allow 5 members to work simultaneously without overwriting each other's work.
CI/CD	Jenkins	To automatically trigger a "build" and run tests whenever code is saved, ensuring the app is always ready to deploy.
Containerization	Docker	To package the app into containers, ensuring it runs exactly the same on our laptops as it does on the cloud server.
Cloud Platform	AWS	To host the application online so it is accessible to job seekers everywhere.
Testing	Selenium	To simulate real users (e.g., clicking "Apply") and verify the website functions correctly.
Monitoring	CloudWatch	To keep an eye on server health and traffic spikes.

5.3 Continuous Integration and Continuous Delivery (CI/CD)

The backbone of our project is the CI/CD pipeline. We established a workflow where code is not just written but continuously tested and delivered.

- Continuous Integration (CI): whenever a team member pushes code to our shared Git repository, Jenkins wakes up. It automatically builds the project and runs our test scripts. This acts as a safety net—errors are caught instantly.
- Continuous Delivery (CD): Once the tests pass, the system is automatically prepared for deployment. This ensures that new features—like a "Report Scam" button—can go live to the public quickly and safely.

Our Pipeline Workflow:

1. Developer commits code to Git.
2. Jenkins triggers a build.
3. Automated tests run to check for errors.
4. Docker creates a secure image of the application.
5. The app is deployed to the AWS Cloud.

5.4 Version Control Strategy

With a team of five members working on different features—such as the user interface, the backend database, and the detection algorithm—coordination was critical. We used Git as our version control system.

Git allowed us to use "branches." For example, one member could work on the *Login System* while another worked on the *Scam Database*, and we could merge these changes later without conflicts. It also gave us a safety net: if a new update broke the site, we could instantly "roll back" to the previous working version.

5.5 Automated Testing

Manual testing is slow and prone to human error. To ensure high quality, we implemented Automated Testing using Selenium.

Selenium acts like a "robot user." It automatically opens our website, types in job URLs, and clicks buttons to ensure everything works as expected. This was crucial for our scam detection engine; we needed to ensure that legitimate jobs weren't accidentally flagged as scams. This automation saved us hours of manual clicking and increased our confidence in the software's stability.

5.6 Cloud Deployment

We chose to deploy the Fake Job & Internship Detection Platform on the cloud (using AWS) rather than a local server. Cloud deployment offers High Availability, meaning the site is up 24/7 for users.

It also provides Scalability. If our platform suddenly goes viral or sees a spike in traffic during internship season, the cloud infrastructure can automatically handle the extra load without crashing. This ensures that job seekers can always access the tool when they need it most.

5.7 Future Enhancements

While the current system is fully functional, we have a roadmap for future improvements:

- AI Integration: We plan to move from keyword detection to Artificial Intelligence (NLP) to detect subtle context in scam emails.
- Browser Extension: A Chrome plugin that automatically highlights fake jobs while users are browsing LinkedIn or Indeed.
- Mobile App: Developing a dedicated mobile application for easier access.
- Community Trust Score: Implementing a system where users can vote on whether a job listing is safe or suspicious.

6. Conclusion

The Fake Job & Internship Detection Platform addresses a critical social issue by leveraging modern technology to protect job seekers. By adopting the Agile methodology, the team was able to prioritize the most important features first.

However, it was the integration of DevOps practices that ensured the system was robust, scalable, and easy to maintain. The automated pipeline allows for continuous improvement of the detection algorithms, ensuring the platform stays ahead of evolving scam tactics. This project demonstrates that combining Agile planning with DevOps execution leads to high-quality, user-centric software solutions.

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